Exercice 2: (8 points)

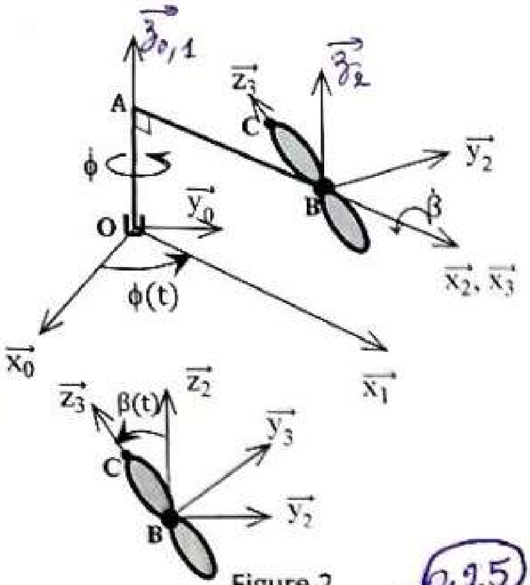

Figure 2

0,25 0,75

1. $\overrightarrow{\Omega^0_3}_{/R2} = \overrightarrow{\Omega^2_3}_{/R2} + \overrightarrow{\Omega^1_2}_{/R2} + \overrightarrow{\Omega^0_1}_{/R2} = \underbrace{\dot\beta \vec{x}_2 + \vec{0} + \dot\Phi \vec{z}_2}_{0,75} = \boxed{{}_{R2}\begin{cases}\dot\beta\\ 0\\ \dot\Phi\end{cases}}$

0,5 0,25 0,25

2. $\boxed{\overrightarrow{V^0(B)}_{/R2} =} \left.\dfrac{d^0\overrightarrow{OB}}{dt}\right|_{/R2} = \left.\dfrac{d^2\overrightarrow{OB}}{dt}\right|_{/R2} + \overrightarrow{\Omega^0_2}_{/R2} \wedge \overrightarrow{OB}_{/R2} = {}_{R2}\begin{cases}0\\0\\0\end{cases} + {}_{R2}\begin{cases}0\\0\\\dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}b\\0\\a\end{cases} = \boxed{{}_{R2}\begin{cases}0\\ b\dot\Phi\\ 0\end{cases}}$

0,5

$\boxed{\overrightarrow{\gamma^0(B)}_{/R2} =} \left.\dfrac{d^0\overrightarrow{V^0(B)}}{dt}\right|_{/R2} = \left.\dfrac{d^2\overrightarrow{V^0(B)}}{dt}\right|_{/R2} + \overrightarrow{\Omega^0_2}_{/R2} \wedge \overrightarrow{V^0(B)}_{/R2}$

$= {}_{R2}\begin{cases}0\\0\\0\end{cases} + {}_{R2}\begin{cases}0\\0\\\dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}0\\ b\dot\Phi\\ 0\end{cases} = \boxed{{}_{R2}\begin{cases}-b\dot\Phi^2\\ 0\\ 0\end{cases}}$ 0,5

0,5 0,5 0,5

3. $\boxed{\overrightarrow{V^0(C)}_{/R2} =} \overrightarrow{V^0(B)}_{/R2} + \overrightarrow{\Omega^0_3}_{/R2} \wedge \overrightarrow{BC}_{/R2} = {}_{R2}\begin{cases}0\\ b\dot\Phi\\ 0\end{cases} + {}_{R2}\begin{cases}\dot\beta\\0\\\dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}0\\ -r\sin\beta\\ r\cos\beta\end{cases} = \boxed{{}_{R2}\begin{cases}\dot\Phi r\sin\beta\\ b\dot\Phi - \dot\beta r\cos\beta\\ -\dot\beta r\sin\beta\end{cases}}$

$\overrightarrow{\gamma^0(C)}_{/R2} = \overrightarrow{\gamma^0(B)}_{/R2} + \left.\dfrac{d^0\overrightarrow{\Omega^0_3}}{dt}\right|_{/R2} \wedge \overrightarrow{BC}_{/R2} + \overrightarrow{\Omega^0_3}_{/R2} \wedge \left(\overrightarrow{\Omega^0_3}_{/R2} \wedge \overrightarrow{BC}_{/R2}\right)$ 0,5

0,5

Avec $\left.\dfrac{d^0\overrightarrow{\Omega^0_3}}{dt}\right|_{/R2} = \left.\dfrac{d^2\overrightarrow{\Omega^0_3}}{dt}\right|_{/R2} + \overrightarrow{\Omega^0_2}_{/R2} \wedge \overrightarrow{\Omega^0_3}_{/R2} = {}_{R2}\begin{cases}0\\0\\0\end{cases} + {}_{R2}\begin{cases}0\\0\\\dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}\dot\beta\\0\\\dot\Phi\end{cases} = \boxed{{}_{R2}\begin{cases}0\\ \dot\beta\dot\Phi\\ 0\end{cases}}$ 0,5

$\left.\dfrac{d^0\overrightarrow{\Omega^0_3}}{dt}\right|_{/R2} \wedge \overrightarrow{BC}_{/R2} = {}_{R2}\begin{cases}0\\ \dot\beta\dot\Phi\\ 0\end{cases} \wedge {}_{R2}\begin{cases}0\\ -r\sin\beta\\ r\cos\beta\end{cases} = \boxed{{}_{R2}\begin{cases}\dot\beta\dot\Phi r\cos\beta\\ 0\\ 0\end{cases}}$ 0,5

$\overrightarrow{\Omega^0_3}_{/R2} \wedge \overrightarrow{BC}_{/R2} = {}_{R2}\begin{cases}\dot\beta\\0\\\dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}0\\ -r\sin\beta\\ r\cos\beta\end{cases} = \boxed{{}_{R2}\begin{cases}\dot\Phi r\sin\beta\\ -\dot\beta r\cos\beta\\ -\dot\beta r\sin\beta\end{cases}}$ 0,5

$$\overrightarrow{\Omega^{0}_{3/R2}} \wedge \left(\overrightarrow{\Omega^{0}_{3/R2}} \wedge \overrightarrow{BC}_{/R2}\right) = {}_{R2}\begin{cases}\dot\beta \\ 0 \\ \dot\Phi\end{cases} \wedge {}_{R2}\begin{cases}\dot\Phi r \sin\beta \\ -\dot\beta r\cos\beta \\ -\dot\beta r \sin\beta\end{cases} = {}_{R2}\begin{cases}\dot\beta\dot\Phi r\cos\beta \\ \dot\Phi^2 r\sin\beta + \dot\beta^2 r\sin\beta \\ -\dot\beta^2 r\cos\beta\end{cases} = \boxed{{}_{R2}\begin{cases}\dot\beta\dot\Phi r\cos\beta \\ (\dot\Phi^2+\dot\beta^2) r\sin\beta \\ -\dot\beta^2 r\cos\beta\end{cases}}$$

0,5

Finalement,

$$\boxed{\overrightarrow{\gamma^0(C)}_{/R2} =} \; {}_{R2}\begin{cases}-b\dot\Phi^2 \\ 0 \\ 0\end{cases} + {}_{R2}\begin{cases}\dot\beta\dot\Phi r\cos\beta \\ 0 \\ 0\end{cases} + {}_{R2}\begin{cases}\dot\beta\dot\Phi r\cos\beta \\ (\dot\Phi^2+\dot\beta^2) r\sin\beta \\ -\dot\beta^2 r\cos\beta\end{cases} = \boxed{{}_{R2}\begin{cases}-b\dot\Phi^2 + 2\dot\beta\dot\Phi r\cos\beta \\ (\dot\Phi^2+\dot\beta^2) r\sin\beta \\ -\dot\beta^2 r\cos\beta\end{cases}}$$

0,5